NanoCGRA-Lite 3x3 OPT — D04 Re-hardened Design Report

Design: output/nanocgra_lite_3x3_opt

Top module: NanoCGRA_Lite

Target template: D04

Date: 2026-08-29

Executive Summary

This report documents the updated nanocgra_lite_3x3_opt submission after re-hardening against the official D04 template. The update fixes the prior block-size audit issue and preserves the original NanoCGRA-Lite functional core.

• Physical block size updated to 550 um x 550 um.

• DEF pin geometry reconciled to the provided D04_D.def template after DBU scaling.

• GDS and filled GDS regenerated with 550 um x 550 um bounding boxes.

• Main DRC, density, antenna, and LVS are passing.

• Original 3x3 CGRA + 32B SRAM + UART functional behavior is preserved inside the D04 wrapper.

Functional Architecture

NanoCGRA-Lite is a compact GF180MCU standard-cell digital accelerator. It contains a 3x3 coarse-grained reconfigurable array, a 32-byte SRAM, a NanoController FSM, and a UART bridge for host command/control.

Block	Description
CGRA	3x3 processing-element mesh, 9 PEs total
SRAM	32-byte, 32x8-bit internal storage
Controller	FSM for command decode, configuration, memory access, and run control
UART bridge	Serial command path using [CMD][ADDR][DATA] packets
D04 wrapper	Physical/template-facing top level that preserves the original core behavior

D04 Wrapper Mapping

D04-facing signal	Internal behavior
clk	Connected to core clk
rst_n	Connected to core rst_n
uart_rx	Connected to core uart_rx
uart_tx_OUT	Driven by core uart_tx
D04 pad-control pins	Tied to fixed safe values
uart_tx_IN	Present for D04 interface compatibility; unused by UART-only core

D04-facing signal	Internal behavior
vdd / vss	Power and ground

Physical Implementation Update

Item	Result
Target slot/template	D04
DEF DBU	2000 DBU/um
DEF die area	(0 0) (1100000 1100000)
Physical block size	550 um x 550 um
Top-level pins	21 D04-facing pins
DEF pin geometry	Matches D04_D.def after 10x DBU scaling
Base GDS bbox	550 um x 550 um
Filled GDS bbox	550 um x 550 um

Signoff Results

Check	Result	Evidence
P&R route check	PASS	logs/pnr_d04.log: 0 pin violations, 0 net violations
Main DRC	PASS	reports/signoff/drc_main_d04_summary.txt: 0 violations
Density	PASS	reports/signoff/density_filled_d04_summary.txt: 0 violations
Antenna	PASS	reports/signoff/antenna_filled_d04_summary.txt: 0 violations
LVS	PASS	reports/lvs/netgen_final.log: Netlists match uniquely
Timing	PASS	logs/pnr_d04.log reports non-negative setup/hold WNS

Final LVS Result

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Circuit 1 contains 5321 devices, Circuit 2 contains 5321 devices.
Circuit 1 contains 5339 nets,    Circuit 2 contains 5339 nets.
Netlists match uniquely.
Result: Circuits match uniquely.
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Key Artifacts

Artifact	Path
Final DEF	pnr/nanocgra_lite_3x3_opt.def
Base GDS	gds/nanocgra_lite_3x3_opt.gds
Filled GDS	gds/nanocgra_lite_3x3_opt_filled.gds
Final LVS log	reports/lvs/netgen_final.log
D04 DRC summary	reports/signoff/drc_main_d04_summary.txt
D04 density summary	reports/signoff/density_filled_d04_summary.txt
D04 antenna summary	reports/signoff/antenna_filled_d04_summary.txt
D04 status	reports/signoff/d04_rerun_status.md

Conclusion

The D04 re-hardened nanocgra_lite_3x3_opt design resolves the audit block-size mismatch while preserving the original registered functional core. The regenerated physical artifacts now align with the official D04 template and pass DRC, density, antenna, and LVS signoff.